The Bombay Salesian Society's

DON BOSCO INSTITUTE OF TECHNOLOGY

Department of Computer Engineering AY 26-27

25CE3CEP01 Community Engagement Project

Class and Div:	Batch:	Group No.:
Comps A	A	3

Group Details:		
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Experiment No.2

Research analysis

Aim: To research the selected problem and explore the current trends, find existing solutions and perform comparative analysis for the same.

Theory:

In Experiment No. 1, we identified a community based problem using empathy tools. In Experiment No. 2, academic papers on the problem topic will be researched to understand the current trends and explored to find existing solutions. The end result of this experiment will be a feature list.

RESEARCH

4-5 research papers on current trends related to your selected problem: Find research papers that discuss how the problem you have selected is being solved currently. List down the methods and features. Provide your findings in table format.

Sr. No	Name of paper	Authors	Year of publication	Methodology	Key Findings	Limitations	References to your work
1	Wi-CaL: WiFi Sensing and Machine Learning Based Device-Free Crowd Counting and Localization	H. Choi, M. Fujimoto, T. Matsui, S. Misaki, K. Yasumoto	2022, IEEE Access, Volume 10,	1.Wi-Fi CSI 2.Collection Environment 3.Variation Detection 4.Noise Removal Feature Extraction RNN Activity Recognition	They specifically investigated changes across different days because factors such as **temperature, humidity and signal interference** can affect the wireless measurements.	Tested only up to 10 people Limited environmental diversity Performance may depend on the room layout Difficulty with very large crowds	Our project is targeting to experiment on **ESP32 node** modules for CSI sensing which is used in the paper.
2	**A Contactless Authentication System Based on WiFi CSI**	Chi Lin , Pengfei Wang,Chuanying Ji Mohammad S. Obaidat (ACM TOSN, 2023)	25 November 2021	Wi-Fi CSI data is collected and analyzed using machine-learning techniques for device-free crowd counting and localization.	The system achieved 92.6% average accuracy across multiple scenes and 97.1% for multiple activities, demonstrating robust CSI-based human sensing.	**Environmental dynamics:** Changes in rooms, corridors, furniture and surroundings can affect CSI signals. **Noise:** Commodity Wi-Fi hardware introduces interference and noise into CSI measurements.	Know UR Crowd is intended for dynamic environments like railway stations and public spaces, the paper's approach to noise reduction and environment-independent feature extraction can help us make our crowd detection system more reliable.
3	**EHUNAM, a WiFi CSI-based dataset for human**	de Armas, E., Diaz, G., Sobron	17 December 2025	Collected Wi-Fi CSI data across multiple environment	Demonstrated multi-person counting (up to 8	The dataset demonstrates multi-person counting, but does not	Provides strong evidence that CSI can support **multi-person sensing**,

	and machine sensing			s, people, machines, and configurations, then used preprocessing and CNN-based classification for human activity and people counting.	people), with **>90% people-counting accuracy in most industrial scenarios**.	address large-scale crowd-density estimation and real-time congestion management in dynamic public spaces.	forming the foundation for our extension from people counting to **crowd-density and congestion detection**.
4	**eHealth CSI: A Wi-Fi CSI Dataset of Human Activities**	Iandra Galdino, Julio C. Soto , Egberto Caballero , Vinicius Ferreira , (Member, Ieee), Taiane Coelho Ramos, Célio Albuquerque , And Débora Muchaluat-Saade , (Member, Ieee)	11 July 2023	1.Wifi Configuration. 2.Human activity collection Participants performed 17 different positions/procedures. 3.Signal processing(Noice removal+Amplitude extraction) 4. Machine Learning	This findings proves that **Wi-Fi CSI can distinguish between an empty and human-occupied environment with very high accuracy under the experimental conditions.**	The study demonstrated up to 99.9% accuracy under controlled conditions, while performance on unseen participants reached 91.18%, highlighting both the potential and the need for real-world adaptation.	We got an idea of model training , testing on data. Wifi-CSI gives fluctuating results in different enviornmrnt, since the research paper aims on detecting motion of humans it targets on more accurate results.
5	WiFi CSI-Based Human Activity Recognition Using Deep Recurrent Neural Network	Jinyang Ding, Young Wong	IEEE Access, 2019,		Instead of directly feeding noisy raw CSI into the classifier, the researchers extract: **CPV** **TFA**	The research focuses on recognizing predefined human activities rather than managing a continuously changing large crowd.	The paper demonstrates that CSI can capture **human activity-related information**,

Link of your research Papers:

1.ContactlessAuthentication- https://dl.acm.org/doi/epdf/10.1145/3532095

2. Wi-CaL: WiFi Sensing and Machine Learning Based Device-Free Crowd Counting and Localization

3. **EHUNAM, a WiFi CSI-based dataset for human and machine sensing**

4. **eHealth CSI: A Wi-Fi CSI Dataset of Human Activities**

5. **WiFi CSI-Based Human Activity Recognition**

3-5 app/website comparisons of already existing complete/partial solutions for your selected problem. List the pros/cons of the existing solutions and write on how the limitations can be improved. Provide your findings in table format.

Sr. No	Name of app/website	Features	Missing Features	References to your work
1	RuView	Camera-free Wi-Fi CSI monitoring; presence/occupancy detection; motion and activity recognition	Limited focus on large-scale crowd management, crowd-density visualization, and venue-level congestion alerts	Helps understand how Wi-Fi CSI can be used for camera-free human sensing
2	ESPectre Sensor	Wi-Fi CSI-based motion detection; works with ESP32; local processing; no camera/audio	Does not directly provide crowd counting, density estimation, zone-wise crowd analysis, or a dedicated crowd-monitoring dashboard	Gives the idea of using Wi-Fi sensing with notifications when unusual activity is detected
3	Xfinity app	Wi-Fi-based motion sensing; live view; adjustable sensitivity; notifications/alerts	Primarily designed for home/security use; lacks large-scale crowd counting, crowd-density analytics, and public-space crowd monitoring	Gives the idea of using Wi-Fi sensing with notifications when unusual activity is detected
4	Congentive	AI-based	Does not	Supports the use of AI for

		analytics and monitoring	specifically provide Wi-Fi-CSI-based device-free crowd sensing and privacy-friendly crowd-density estimation	analysing sensing data and generating useful insights
5	**Wavey**	Wi-Fi CSI-based occupancy/motion sensing; camera-free sensing; real-time sensing	Limited dedicated functionality for large-scale crowd-density estimation, congestion prediction, crowd alerts, and crowd-management dashboards	Provides a reference for using Wi-Fi CSI for occupancy and movement detection

Based on your findings, now list down the features that will be there in your solution and justify their need

Name of app/website: Know UR Crowd		
Sr. No	Features	Justification
1	Alert Based Systems	Based on the sudden changes/movements detection .It will give emergency alerts / raise awake calls.
2	Zone-wise monitoring	Setting-up the wifi modules in physically accurate method in accord to et high amplitude/frequency match.Dividing a particular space in zone .
3	Density classification	Crow density classification.(eg.Red,Green,Orange category)